Day 1 – Introduction to OOP

Daily Time: 2 Hours | Level: Foundation → Intermediate

Focus: Understand Class & Object deeply

Today you stop writing random code.

Today you start building structure.

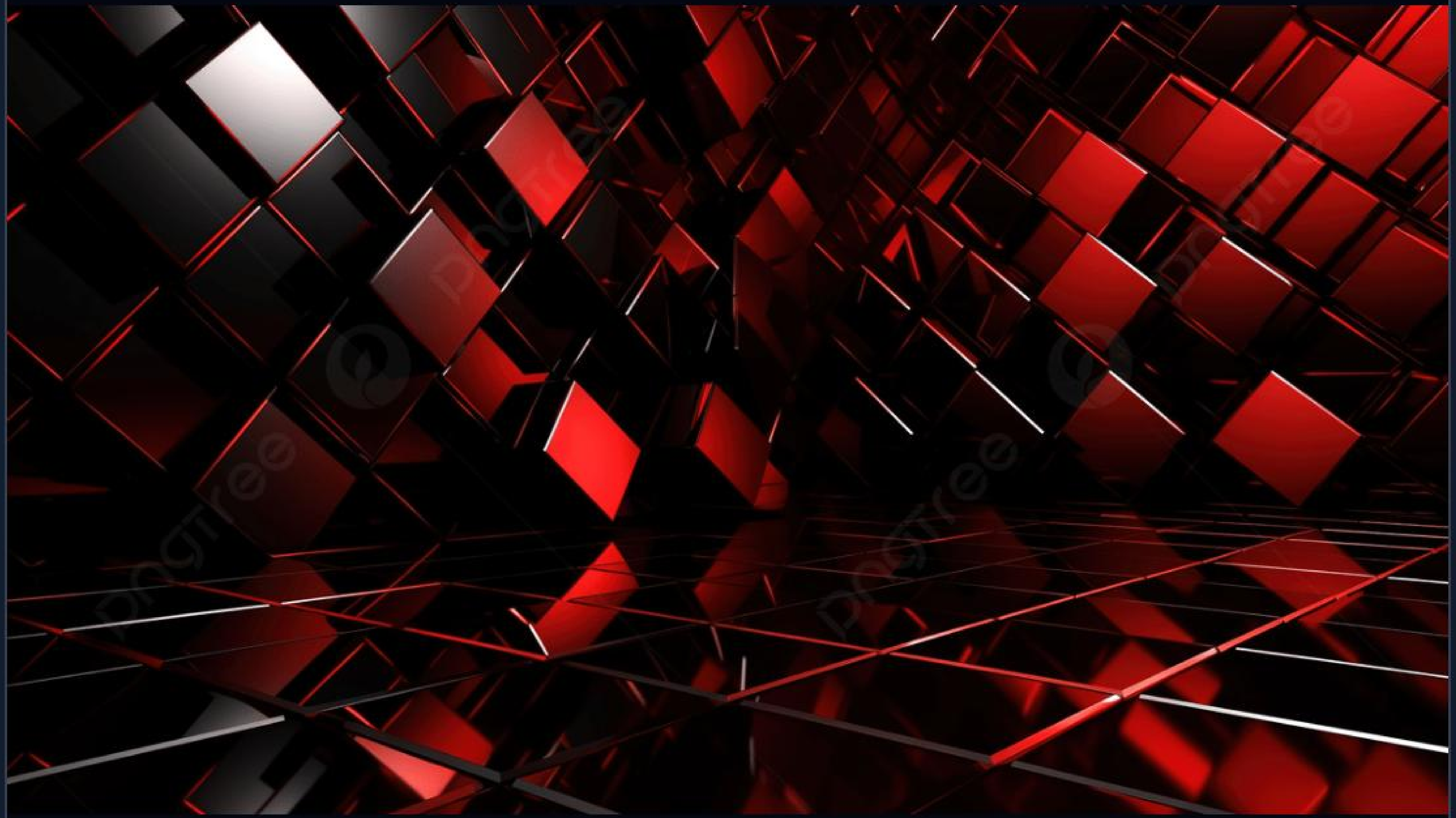
What is OOP?

OOP = Object Oriented Programming

It is a way of writing programs using:

- Objects
- Classes
- Real-world modeling

> Instead of random functions, we create blueprints and build objects.



Real-Life Example: Think of a Car

Attributes & Methods

Car has:

- color
- brand
- speed

Car can:

- start()
- stop()
- accelerate()

The OOP Mapping

Car = **Class**

Your specific car = **Object**

A class defines the properties and behaviors, while an object is a specific realization of that class.

What is a Class?

A class is a **blueprint**.

```python
class Car:
    pass
```

Right now this class does nothing.

But it defines structure.

What is an Object?

An object is a **real instance** created from a class.

```
my_car = Car()
```

Now `my_car` is an object of class `Car`.

First Simple Example

Python Code

```python
class Student:
    pass

s1 = Student()
print(s1)
```

Console Output

```
<__main__.Student object at 0x7f8b9c ... >
```

This means the object is created successfully in memory.

Hour 2 – Attributes

Now we add data inside the class.

We use a **constructor**.

Constructor (__init__)

The Code

Runs automatically when object is created.

```python
class Student:
    def __init__(self, name, age):
        self.name = name
        self.age = age

s1 = Student("Saurav", 21)
```

Accessing Attributes

```python
print(s1.name)
print(s1.age)

# Output:
# Saurav
# 21
```

Understanding self

> self refers to the current object. When we write:
>
> `self.name = name`
>
> It means: Object’s name = given name.

— Each object keeps its own data securely.

Creating Multiple Objects

Instantiation

```python
s1 = Student("Amit", 20)
s2 = Student("Rahul", 22)
```

Independence

```python
print(s1.name) # Amit
print(s2.name) # Rahul
```

🔒 Each object stores different data independently.

Output Prediction 🧪

What is the output of the following code?

```
class Car:
    def __init__(self, brand):
        self.brand = brand

c1 = Car("BMW")
c2 = Car("Tesla")

print(c1.brand)
print(c2.brand)
```

CORRECT OUTPUT:

```
BMW
Tesla
```

Practice Task 1: Person

Create a class **Person** with:

- `name`
- `age`

>_ Create 2 objects and print details.

Practice Task 2: Book

Create a class **Book** with:

- `title`
- `author`
- `price`

>_ Create 1 object and print values.

Practice Task 3: Mobile

Create a class **Mobile** with:

- `brand`
- `model`

> Create 3 different mobiles.

Final Practice Task: Animal

Create a class **Animal** with:

- `name`
- `type`

>_ Print animal details.

Think Like a Developer 💡

The Core Question

Ask yourself:

"Why use classes instead of normal variables?"

The Reality of Scale

Because in large software systems:

- Users are objects
- Products are objects
- Orders are objects
- Payments are objects

Everything is modeled as objects. That’s how real software works.

Day 1 Summary

Core Concepts

Successfully understood the fundamentals of OOP, including the definitions of **Class** and **Object**.

Mechanisms

Learned how a **constructor** initializes data and deeply understood the purpose of **self**.

Architecture

Created multiple independent objects. You are no longer just writing scripts; you are learning **software architecture**.

Image Sources

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